MICHAEL MILOWSKI

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EDUCATION

University of Illinois Urbana-Champaign **August 2024 – May 2028**
Bachelor of Science in Aerospace Engineering GPA 3.76/4.00

EXPERIENCE

NASA Langley Research Center – Space Mission Analysis Branch, Pathways Intern **August 2026 – Present**

- Analyzing mission architectures and technical parameters to support cost estimates, feasibility assessments, and leadership decisions.
- Conducting research to compare SEER cost estimates with historical mission data to evaluate model accuracy and inform NASA cost-estimating best practices.
- Completed a NASA systems engineering course delivered through Caltech, covering requirements development, technical reviews, risk management, verification, and validation.

Woodward Inc. – Aerospace Electronics Systems NPI Engineering Intern **May 2026 – August 2026**

- Troubleshot and integrated an electro-mechanical thrust reverser actuation system, doing PCB rework and repair, building and reworking cables and performing Acceptance Test Procedure.
- Reviewed and corrected system requirements for a military actuation system to ensure alignment with customer specifications.
- Analyzed components and assembly weight for complex electronics housing to improve weight estimation.
- Designed and 3D printed tooling to greatly improve efficiency and accuracy of resolver testing prior to installation.

PROJECTS AND LEADERSHIP

Illinois Space Society – NASA Human Lander Challenge (HuLC) **August 2024 – May 2026**

- Analyzed cryogenic propellant transfer methods to improve pipe chilldown efficiency by up to 176% and reduce propellant loss by 65%.
- Conducted literature reviews of 30+ research papers and applied findings to Artemis propellant transfer conditions.
- Presented concept to a NASA Glenn cryogenics engineer and was selected as one of 12 HuLC competition finalists.

NASA L'SPACE – Deputy Project Manager of Resources **January 2025 – May 2025**

- Led weekly meetings to coordinate goals, communicate progress, and plan key programmatic elements of the mission, while fostering open communication and problem-solving throughout a diverse multidisciplinary team.
- Managed mission cost, schedule, and risk while using NASA mission-planning tools and life-cycle processes to develop a space mission concept through Preliminary Design Review.

Kappa Theta Tau – Professional Society for Distinguished Engineering Students **September 2024 – Present**

- Contributed to the 2025 University of Illinois Engineering Open House in designing, prototyping and completing a project, strengthening my knowledge of the engineering process through collaborating with upperclassmen.
- Collaborated with a team of peers to plan, organize, and execute three successful fundraising events raising $1000+ for charity while developing effective communication and project management skills.
- Orchestrated a student STEM program at Champaign Public Library to promote engineering in a fun and informative way by developing and leading a lesson and activity for children in afterschool care.

AWARDS

Best Technical Presentation – NASA HuLC Team
Dean's List - University of Illinois Urbana-Champaign
Magna Cum Laude Graduate, GPA: 4.5/4.0 – Lake Zurich High School
AP Scholar with Distinction – Highest AP award for excellence across 8 exams

SKILLS

CAD & Design Software: Autodesk Inventor, AutoCAD, Siemens NX, Fusion 360, Creo
Programming Languages: C++, Python, Java
Workspace Tools: Excel, IBM DOORS, Jira, Windchill, Microsoft 365, SEER